A LOCAL-FINITENESS OBSTRUCTION TO THE REGULARITY PREMISE IN FU'S THEOREM 2.8

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Abstract. In a 2026 manuscript investigating trans-exponential o-minimal expansions of the real ordered field, Theorem 2.8 assumes that for every allowed polynomial-exponential system P, the regular parameter locus $R \subset G \times [-1,1]^{2n+2}$ admits a triangulation by simplices $\{\Delta_\alpha\}$ that is locally finite in the full ambient parameter space $G \times [-1,1]^{2n+2}$. We prove that for the allowed system $n = 1,\ k = u = 0,\ P(x, l_1, l_2, \epsilon, \delta) = x^2$, the identity-coordinate slice of R in $[-1,1]$ is $[-1,1] \setminus \{0\}$, which is not closed. Because the union of any locally finite family of closed subsets in a topological space is closed, no locally finite triangulation of R by closed simplices exists in $G \times [-1,1]^{2n+2}$. If the simplices Δ_α are instead taken to be open simplex interiors, then the closedness step in the subsequent Claim 2.9 fails. The topological and logical core is verified in Lean 4.

1. Main Obstruction Theorem

Let $n = 1$ and $k = u = 0$, and consider the allowed polynomial system

$$P(x, l_1, l_2, \epsilon, \delta) \;=\; x^2.$$

For $(A, \eta, l, \epsilon, \delta) \in G \times [-1,1]^4$, the regular parameter set is defined by

$$R \;=\; \big\{(A, \eta, l, \epsilon, \delta) \in G \times [-1,1]^4 : P \circ A - \eta \text{ has nonsingular zeros only}\big\}.$$

Fixing the identity affine transformation $A = \mathrm{id}$ and auxiliary parameters $(l_1, l_2, \epsilon, \delta) = (0,0,0,0)$, the univariate polynomial $x^2 - \eta$ has a singular zero at $x = 0$ when $\eta = 0$ and only nonsingular real zeros (or no real zeros) when $\eta \in [-1,1] \setminus \{0\}$.

Theorem 1.1. *For $P(x, l_1, l_2, \epsilon, \delta) = x^2$, the slice of R along the identity coordinate $A = \mathrm{id}$ and $(l_1, l_2, \epsilon, \delta) = 0$ is*

$$R_{\mathrm{id}} \;=\; [-1,1] \setminus \{0\}.$$

Consequently:

(1) *R is not closed in $G \times [-1,1]^4$.*
(2) *There exists no family $\{\Delta_\alpha\}$ of closed simplices with $\bigcup_\alpha \Delta_\alpha = R$ that is locally finite at every point of $G \times [-1,1]^4$.*
(3) *If $\{\Delta_\alpha\}$ is instead a locally finite family of open simplices with $\bigcup_\alpha \Delta_\alpha = R$, then individual members Δ_α are not closed in $G \times [-1,1]^4$, obstructing the closed-membership inference in Claim 2.9.*

Proof. Since $0 \in \overline{[-1,1] \setminus \{0\}} \setminus R_{\mathrm{id}}$, R_{id} is not closed in $[-1,1]$, and by continuity of the slice inclusion $\eta \mapsto (\mathrm{id}, \eta, 0, 0, 0, 0)$, R is not closed in $G \times [-1,1]^4$. In any topological space X, if $\{F_\alpha\}$ is a locally finite collection of closed sets, then $\bigcup_\alpha F_\alpha$ is closed in X; applying this to $X = G \times [-1,1]^4$ proves (2), and (3) is immediate. □

Remark 1.2 (Scope Boundary)**.** Theorem 1.1 identifies a gap in the stated regularity premise of Theorem 2.8 and its application in Claim 2.9. It does not prove or refute the existence of a trans-exponential o-minimal expansion of $(\mathbb{R}, <, +, \cdot, \exp)$.

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2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1465 in the domain of model theory, o-minimal structures, and real analytic geometry follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *proof premise correction* deciding the scope: closed-simplex local-finiteness premise in Theorem 2.8 of arXiv:2604.03477v2 for P(x)=x^{}2.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- existence of a trans-exponential o-minimal expansion of the real ordered field.

References

[1] Primary problem list and source literature for Result 1465 (model theory, o-minimal structures, and real analytic geometry), archived with exact theorem locators and verification certificates in the accompanying repository package.

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